Exact-information chain: state, count attachment and births

7 October 2026 — G1 approved; insertion contract Q-E unselected

Scope. Source transcription and reviewer deductions. Every distinct restriction may be retained. No compression/cost criterion, new approximate recovery closure, traffic simulation or useful-gradient claim. A complete joined birth/node solver and clearance remain unqualified.

1. Retained state and conservative population

At a declared update phase, keep the own traffic field U, speed-support history $\mathcal{H}$, waiting inventories W, accepted transfer counters H_e and all restriction records:

$$X = (U, \mathcal{H}, W, \{H_e\}, \{z_j\}), \qquad z_j = (\ell_j, x_j, v_j, v_{\mathrm{des},j}, \theta_j, h_j, e_j, t_j^{\mathrm{b}}). \tag{1}$$

Here ℓ_j is lane; x_j position (m); $v_j, v_{\mathrm{des},j}$ realized/desired speed (m/s), retaining the phase-required value; θ_j contains fixed acceleration/type/grade data; h_j is the lane-count association (vehicle equivalents); e_j is birth-stream metadata; t_j^{b} birth time (s). U contains the spatial mass/count information needed by the declared KW method. $\mathcal{H}$ contains own preceding-period information needed by source lane-choice speeds, initialized publicly. None is a reference-future oracle.

For cell/volume c, mass n_c (vehicle equivalents) and accepted transfer mass F_{cb}, the paired update is

$$n_c^+ = n_c + \sum_b F_{bc} - \sum_b F_{cb}, \qquad M^+ - M = I - E. \tag{2}$$

M includes all waiting, ramp and road mass; I, E are external arrival/exit increments. All internal transfers cancel. Descriptors add no mass. A source lateral transfer already advances to the downstream target section.

For upstream ramp permission r^k (veh/s), offered mass A^k, waiting mass W_r^k, approach receiving R_a^k (veh/s) and step Δt (s),

$$B^k = \min\{W_r^k + A^k, r^k \Delta t, R_a^k \Delta t\}, \quad W_r^{k+1} = W_r^k + A^k - B^k. \tag{3}$$

This admitted mass enters the approach, not directly the freeway merge. At the approved point node let D_r, D_m, R be ramp/non-ramp demands and shared receiving (veh/s), and $p \in (0,1)$ fixed ramp priority:

$$(f_r, f_m) = \begin{cases} (D_r, D_m), & D_r + D_m \le R, \\ (\min\{D_r, \max\{pR, R - D_m\}\}, R - f_r), & D_r + D_m > R. \end{cases} \tag{4}$$

The first component is evaluated before the second. Existing coupled restrictions must determine R consistently; this formula does not supply it. These are previously approved interfaces, not a new reference law.

2. Exact source correction to a moving traffic label

Reviewer derivation. On a regular lane interval with fixed upstream boundary x_0, let ρ be density (veh/m), q longitudinal flow (veh/s) and s net lateral source (veh/(m s)). Conservation is

$$\partial_t \rho + \partial_x q = s. \tag{5}$$

Let $C(t)$ be accepted cumulative inflow at x_0, with $\dot{C} = q(t, x_0)$. Choose one fixed label gauge and define

$$N(t, x) = C(t) - \int_{x_0}^{x} \rho(t, y)\, dy. \tag{6}$$

Differentiation and conservation give

$$N_x = -\rho, \tag{7}$$

$$N_t = q(t, x) - \int_{x_0}^{x} s(t, y)\, dy. \tag{8}$$

For a path $x = \phi(t)$ with speed $v = \dot{\phi}$, set $h(t) = N(t, \phi(t))$. Thus

$$\dot{h} = q(t, \phi) - v\rho(t, \phi) - \int_{x_0}^{\phi} s(t, y)\, dy. \tag{9}$$

Where the embedded zero-passing condition gives zero relative flux $q - v\rho = 0$, this becomes

$$\dot{h} = -\int_{x_0}^{\phi} s(t, y)\, dy. \tag{10}$$

If $s = 0$, h is constant. With lateral entries/exits upstream of the path, its label changes despite zero passing. This is a coordinate/count correction, not an extra service penalty or population term.

Discrete fixed-section version. For population P_x between x_0 and fixed x, accepted boundary mass ΔC, crossing mass F_x and net lateral addition L_x in that section,

$$\Delta P_x = \Delta C - F_x + L_x, \qquad \Delta N_x = \Delta C - \Delta P_x = F_x - L_x. \tag{11}$$

An upstream net lateral entry of one unit shifts N by -1 when $F_x = 0$. An exit shifts it by $+1$. Boundary admissions also increment C and so cancel their inventory contribution in this coordinate. Weak/discrete balances extend the bookkeeping beyond smooth fields; they do not choose which side owns a point source exactly on a path. That attachment remains a joined-law obligation.

At a point merge chosen as x_0, the boundary ledger includes both accepted incoming streams. It is not a lateral source inside (x_0, x).

3. Motion, birth clocks and count attachment

Source motion. At the source sample phase, motion obeys

$$v_j(t) = \min\{v_{\mathrm{dwn},j}(t), v_{\mathrm{des},j}(t)\}, \tag{12}$$

$$v_{\mathrm{des},j}(t+\Delta t) = v_j(t) + \left[a_{0,j}\left(1 - \frac{v_j(t)}{v_{\max,j}}\right) - gG(x_j(t))\right]\Delta t, \tag{13}$$

$$x_j(t+\Delta t) = x_j(t) + v_j(t)\Delta t. \tag{14}$$

v_{dwn} is own coupled immediately downstream KW speed; a_0 is zero-speed acceleration ($\mathrm{m/s^2}$), $v_{\max}$ maximum speed, g gravity and G dimensionless grade. Lane-choice look-ahead speed is a different quantity. Exact retention does not justify assigning downstream speed independently.

For generating stream e, define public progress $\eta_e \in [0,1)$ and actual accepted rate f_e:

$$H_e(t) = \eta_e + \int_0^t f_e(s)\, ds. \tag{15}$$

The sampled source birth count at update t_k is

$$\Delta K_e^k = \lfloor H_e(t_k)\rfloor - \lfloor H_e(t_{k-1})\rfloor. \tag{16}$$

No conserved quantity is rounded. The proposed E1 reset adds these records after the completed transfer update, with declared placement/speed/count data. It does not retroactively alter completed transfers.

E2 instead proposes each continuous crossing time

$$\tau_{e,m} = \inf\{t : H_e(t) \geq m\}, \qquad m = 1, 2, \ldots. \tag{17}$$

It requires coupled continuation after τ, because the new restriction may alter the remaining f_e. It explicitly reopens the selected sampled clock and dependent onset/inactive classification if approved; neither is reopened now. It is an extension, not a source theorem.

Clock input example only. Give $H(t) = 3/4 + t/2$ on $[0,1]$ (veh) and a sampled update at $t = 1$ s. Then

$$\tau_1 = \tfrac{1}{2}\ \mathrm{s}, \qquad \Delta K = 1, \tag{18}$$

whereas the sampled record starts at 1 s. No joined traffic solution, feasible common-control pair or TST difference is asserted from this prescribed counter.

The birth count association must match the own traffic field at its location and phase. At a point ramp boundary it uses total accepted mainline plus ramp count in the same gauge. At an internal point source, one-sided attachment must be explicit. Neither clock choice alone certifies simultaneous events.

4. Conditional equivalence and full-system-time obligations

Suppose the complete joined update $\mathcal{T}_r$ is well defined and invariant under descriptor renaming. Let Π be a lossless representation with the independent own update $\widetilde{\mathcal{T}}_r$, and assume

$$\Pi\mathcal{T}_r(X) = \widetilde{\mathcal{T}}_r(\Pi X). \tag{19}$$

Starting from equivalent public states, induction gives equivalent states and exits at every common update. Keeping all distinct complete records can meet the information part of this identity. The identity does not prove existence of $\mathcal{T}_r$ or fill an unspecified birth/reset. It also does not prove that personal identities or every retained record are necessary.

Let M_0 be initial total mass, $A(t)$ cumulative offered arrivals and $E(t)$ cumulative sink exits. Include all outside waiting and finite ramp/road mass:

$$M(t) = M_0 + A(t) - E(t), \qquad J = \int_0^{T_{\text{clear}}} M(t)\,dt. \tag{20}$$

J has units veh s. For two schedules a, b with common population and arrival history, extend cleared exits constantly to common T:

$$J^a - J^b = -\int_0^T [E^a(t) - E^b(t)]\,dt. \tag{21}$$

Equivalent complete exit histories imply equal TST, including relocated queues. Finite-horizon equality is insufficient if remaining tails differ. An integrable certified inventory tail may replace finite exact clearance. Positive final permission and finite population alone do not prove that tail for this joined law.

Disposition. Exact-state own evolution has source-backed ingredients and source-correct count interfaces. The complete birth/activation/reset contract is Q-E; E1/E2 are unselected. Joined node/coincidence solution, clearance, nontrivial full-TST rate leverage, reduced-model claims and useful gradients remain unqualified. No exact-state impossibility or necessity of approximation is inferred. Stage 3 open, Stage 4 and experiments unopened.

Source locations. Supplied Laval/Daganzo (2006), Section 2 and Appendix A; 2004 antecedent Section 3; Daganzo/Laval RR-2003-2 Sections 2–4/Appendix; Daganzo PWP-94-12 Section 3.2. Source-correct count identities, equivalence induction and the clock example are reviewer deductions. Versions, URLs and transfer limits are in `exact_information_source_audit.md`.